



Largest 3D map of the Universe

The map includes over 2 million galaxies and covers a whopping 11 billion years of the universe's history. <https://digit.in/aug20-01>



Spiral galaxy's magnetic field

The image shows a galaxy called NGC 4217, a distant spiral galaxy, and its complex magnetic field. <https://digit.in/aug20-02>

HOT NEPTUNE DESERT

Extrasolar planets the size of Neptune in close orbits around the host star are rare, and astronomers might have finally found out why

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Credit: MIT/NASA/JPL/Caltech

In 2001: A Space Odyssey, a race of advanced extraterrestrials use a cosmic equivalent of a Swiss army knife, to transform the entire planet of Jupiter. Unlike the very human visions of terraforming a planet to make it more like the Earth, the effort of the fictional aliens was to turn Jupiter into a star. This would make for a binary system, allowing life of any kind to spring up on the Jovian moons, particularly on Europa. Such an engineering project would require changing the mass and density of Jupiter.

This is, of course fiction, and actually fiddling around with the masses of planets can cause all kinds of disastrous situations, similar to the early period of the Solar System which was marked by cataclysmic collision events between protoplanets, the effects of which are seen to this day. It would disturb the orderly orbits of the planets in the system. The increase in mass would mean that the gas

giant would drift closer to the Sun, mostly going into an elliptical orbit that would periodically expose the world to the extreme energies of the Sun. For both the gas giants as well as the ice giants of Uranus and Neptune, getting closer to the Sun would strip away the material that form their dense atmospheres, and liquid oceans. These jets of material being stripped away from the planet would be much more spectacular than the tails we see of comets. The planet would rapidly lose mass, and spiral inwards into the Sun, ultimately getting destroyed. In rare cases, they may remain in stable orbits, much more massive than the Earth, but just a shadow of the giants that they once were. If you want to see this process in action, the Universe Sandbox is a great tool that allows you to perform these simulations on your computer. In many star systems, a gas giant such as Jupiter is not always located in the outermost parts of the system. They can be much closer to the host star, and belong to a class of planets called Hot Jupiters.

One of the most popular and effective methods of detecting exoplanets is the transit method, which looks for telltale dips of light by watching the same star over long periods of time. This involves detecting the amount of light blocked from the host star, by a planet passing in front of it as seen from the Earth, millions of light years away. From these tiny fluctuations, scientists can derive the masses of the stars, as well as their orbital periods. Naturally, the planets that are closer to the star, and the ones with smaller years, are the planets that are most detected. For years, astronomers were puzzled by what they called desert of Hot Neptunes. Planets in close orbits around the host star ten to twenty times the size of the earth are called Hot Neptunes, while the even larger gas giants ten to twenty times the size of Neptune are called Hot Jupiters. While there were plenty of smaller Earth-sized rocky exoplanets, as well as massive Jupiter-sized gas giants, there was a dearth of